

Ankit Ghosh

SIGNAL & IMAGE PROCESSING · MACHINE LEARNING · INDUSTRIAL AI · ACOUSTIC SENSING

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“Approaches AI as a connected system in which physical signals, process knowledge, data architecture, deployment constraints and operator workflows connect through shared domain rules.”

Summary

Machine learning engineer and researcher working on the recovery of structure from noisy physical measurements: acoustic, vibration, optical and time-series. My doctorate built aMP, an end-to-end deep learning pipeline that replaces manual staging of meiosis in live-cell confocal microscopy, combining segmentation-based motion stabilisation, ensemble object localisation, constrained tracking and state-vector sequence modelling. Earlier work was in underwater acoustics: channel estimation in a water column, chirp-modulated communication, and passive SONAR source separation. I now lead industrial AI at BATO GmbH, owning platform architecture, on-premise deployment and technical delivery to manufacturing customers. I have worked in research and industry in parallel since 2015, including in regulated environments where correctness is defined by a standard text and an audit trail rather than by a validation metric alone. Python and PyTorch daily; C++ in production at DNV and Scalian; German at B2.

Work Experience

BATO GmbH (spun out of Hummingbird Converting Solutions)

Hamburg, Germany

TECHNICAL LEAD, INDUSTRIAL AI — Machine Learning Engineer

Feb. 2025 - Present

- Set the reference architecture for BATO's AI products and own it end to end — model and inference topology, data layer, deployment boundary and security model — then build against it as primary engineer.
- Shipped BATO KI Interview**, a German-first, voice-enabled LLM system, to a private pilot at a paper manufacturer. Experienced machine operators are interviewed by voice or text; the system transcribes locally, grounds every turn in approved machine documents and the running interview, and generates adaptive follow-up questions to capture troubleshooting knowledge that exists in no document.
- Designed the system so the model proposes and deterministic logic decides: the LLM generates candidate questions, an explicit **scoring and gating layer** selects the next one, participant knowledge is **isolated between interviews**, and every generated claim carries person and interview **provenance**.
- Built the human-in-the-loop controls that make it deployable: consent flow, participant correction of transcripts, explicit release of the final summary before anything is stored, a review gate before an uploaded document becomes retrievable, and readiness checks that refuse interviews when storage or inference is unavailable.
- Implemented the full stack for it — LLM orchestration, hybrid dense and lexical **retrieval**, multilingual speech-to-text and text-to-speech, PostgreSQL as system of record with a Neo4j knowledge-graph projection, and the administrative interfaces — and deployed it on **private GPU infrastructure with local model inference** and invitation-only access, so operator data never leaves the controlled environment.
- Own the architecture of **BATO Verse**, a tenant-aware industrial knowledge platform (FastAPI, PostgreSQL/pgvector, Neo4j, TimescaleDB, MinIO, Docker), whose retrieval workflows link time-series anomalies to technical manuals and operator knowledge to identify probable fault sources and suspect sensors.
- Built a production copilot for a six-colour rotogravure printing line, combining machine settings, quality parameters, sensor data and deterministic process knowledge into operator recommendations.
- Design secure on-premise architectures with read-only PLC/SCADA integration, local inference and isolated storage — no external cloud dependency for production data — and produce the hardware-sizing and security-boundary documentation customer IT and works councils require before deployment.

Aidy Health

Munich, Germany

AI CONSULTANT (part-time engagement)

Jun. 2025 - Oct. 2025

- Advised on signal processing for WiFi **Channel State Information (CSI)** data, targeting accuracy and real-time sensing performance.
- Developed models to detect and classify objects under varying motion and material conditions for passive environmental sensing.

Metric-space (AI division, spun out of PANDA GmbH)

Hamburg, Germany

MACHINE / DEEP LEARNING ENGINEER

Sep. 2021 - Sep. 2024

- Built and deployed **vibroacoustic predictive maintenance** (v1.0) for rotating machinery, from raw sensor data through model development to production release.
- Developed **real-time acoustic event detection** for vehicle tampering on Jetson Xavier NX, using wavelet-decomposition preprocessing to meet latency targets on edge hardware.
- Designed an unsupervised forecasting and clustering method combining Meta's **Encodec** with the **PHATE** algorithm, reducing inference latency on large-scale time-series and surfacing latent regime structure.
- Built an unsupervised clustering and labelling toolbox for **equine gait analysis**, classifying movement anomalies across multi-view video.
- Deployed ensemble adaptive-threshold **anomaly detection on infrared imagery** using the CFLOW anomaly foundation model.
- Integrated **OpenAI function calling** into a conversational interface over time-series datasets, letting analysts query and interrogate data directly.

Scalian

DATABASE APPLICATION DEVELOPER (working student, alongside doctoral research)

Hamburg, Germany

Feb. 2020 - Feb. 2021

Migrated an Airbus finance model from PowerPivot to Python, converting spreadsheet-embedded logic into a maintainable, version-controlled codebase, and maintained it in service.

Contributed to an in-house ERP application (C++, ASP.NET).

DNV - Renewable Advisory

SOFTWARE DEVELOPER & SIGNAL ANALYST

Hamburg, Germany

Aug. 2017 - Feb. 2019

Analysed **harmonics, inter-harmonics and high-frequency content** in raw grid measurements for synchronous grid operation, and extended flicker analysis to current recommendations.

Improved over- and under-voltage detection algorithms against defined standard thresholds.

Implemented several **IEC standards** as verified applications with automated reporting and plotting (C++, VB.NET).

Basler AG

MACHINE VISION PLACEMENT — Fishing for Experience university–industry programme

Ahrensburg, Germany

Nov. 2015 - Mar. 2016

Designed, built and installed a registered-mark (passer-mark) inspection system for **colour shift control** on a printing line, including a timer interface synchronising mark detection to press motion.

Education

Universität Hamburg

PH.D. IN MACHINE LEARNING AND IMAGE PROCESSING

Hamburg, Germany

Jan. 2019 - Oct. 2024

Grade: cum laude (1.7). Advisor: Prof. Dr. Arp Schnittger. Second examiner: Prof. Dr. Sören Laue. Funded by the **Human Frontier Science Program (HFSP)**.

Thesis: *aMP: Deep learning framework to characterize meiosis progression timeline in tetraploid Arabidopsis thaliana*. Built an end-to-end pipeline replacing manual staging of meiosis from confocal live-cell imaging: anther segmentation and motion stabilisation, meiocyte localisation, tracking, and stage classification.

Segmentation and stabilisation of anthers via affine transformation to hold the focal plane across frames — Dice 88.9%, IoU 80.8%, F1 88.9% on the held-out test set.

Meiocyte localisation using an ensemble of EfficientDet backbones with non-maximum suppression, evaluated under the COCO protocol — precision 0.77, recall 0.79, F1 0.78.

Tracking through a restricted-space motion model, associating detections across frames on normalised cross-correlation and IoU, reducing computation time against unconstrained association.

Represented meiosis phases as **state vectors**, validated predicted sequences against a knowledge graph, and derived Z-normalised staging pathways to build a piece-wise meiotic timeline. Per-stage classification accuracy 76–91%.

Applied the timeline to **tetraploid A. thaliana**: meiosis I comparable to diploid, meiosis II markedly prolonged; quantified temporal dynamics in txc5;6 mutants and ATM insertions.

Reused the pipeline’s modules for further quantification tasks — DNA double-strand-break foci, BiFC analysis, pollen counting.

Continued as **postdoctoral researcher** (Oct. 2024 - Feb. 2025) on unsupervised staging of maize meiocytes.

Hamburg University of Technology (TUHH)

M.Sc. INFORMATION AND COMMUNICATION SYSTEMS

Hamburg, Germany

Oct. 2015 - Sep. 2018

Overall grade 1.7. Major: communication systems.

Master’s thesis: *Source Separation and Echo Detection of Underwater Acoustic Signals of Whales* (grade 1.3), Helmut Schmidt Universität, Jan. - Jun. 2018, under Prof. Dr.-Ing. habil. Udo Zölzer.

Project: chirp modulation for resilient underwater acoustic communication, Institute smartPORT, under Prof. Bernd-Christian Renner.

Previously studied M.Sc. Microsystems at Bremen University of Applied Sciences (Oct. 2014 - Sep. 2015) before transferring; coursework average 1.4, project on **channel state estimation in a water column** under Prof. Dr. Dieter Kraus.

Institute of Engineering and Management

B.TECH. ELECTRONICS AND COMMUNICATION ENGINEERING

Kolkata, India

Aug. 2010 - May 2014

DGPA 8.3.

Skills

Programming	Python, C++, C, MATLAB, SQL
Machine Learning	PyTorch, TensorFlow, Keras, NumPy
LLM & GenAI Systems	Local/self-hosted inference, RAG & hybrid retrieval, LLM orchestration, tool & function calling, speech-to-text and text-to-speech, knowledge graphs, output gating & provenance, human-in-the-loop review
Methods	Signal & time-series processing, acoustic signal processing, image segmentation & tracking, anomaly detection, foundation-model adaptation, unsupervised clustering
Architecture & Infrastructure	FastAPI, PostgreSQL/pgvector, Neo4j, TimescaleDB, InfluxDB, MinIO, Docker, on-premise GPU deployment, edge inference (Jetson), read-only PLC/SCADA integration
Languages	English (Native/Bilingual), Bengali (Native), Hindi, German (B2)

SEPTEMBER 7, 2026

ANKIT GHOSH · RÉSUMÉ

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Research Projects

Helmut Schmidt Universität (Universität der Bundeswehr Hamburg)

Hamburg, Germany

MASTER'S THESIS — PASSIVE ACOUSTIC MONITORING & BIOACOUSTICS

Jan. 2018 - Jun. 2018

- *Source Separation and Echo Detection of Underwater Acoustic Signals of Whales*. Grade 1.3. Supervisor: Carlos de Obaldía, M.Sc. Examiners: Prof. Dr.-Ing. habil. Udo Zölzer, Prof. Dr. Bernd-Christian Renner.
- Built a system to detect **echolocation clicks of toothed whales** and extract three features from them sufficient to identify species, working from passive acoustic monitoring recordings containing multiple species and ambient ship noise.
- Isolated clicks from whistles, creaks and other harmonic vocalisations using **generalised Wiener-filtering source separation** — non-negative matrix factorization, median filtering and kernel additive modelling — compared by peak-to-noise ratio and signal-to-distortion ratio.
- Derived **echo decay time** via Hilbert transform analysis, establishing that each click pulse carries superimposed pulses.
- Separated each pulse into **high- and low-energy components** to detect the inlying short-term pulses and identify the frequency bands they occupy, using continuous wavelet analysis.
- Estimated **click length** through an adapted correlation technique exploiting the band-limitation of each pulse.
- Modelled shallow-water channel behaviour with synthetic impulse responses to reason about echo content when the source location is unknown.

Research group smartPORT, Hamburg University of Technology

Hamburg, Germany

MASTER'S PROJECT — UNDERWATER ACOUSTIC COMMUNICATION

Nov. 2016 - Jun. 2017

- *Chirp Modulation for Resilient Underwater Acoustic Communication*. Grade 2.0. Supervisor and first examiner: Prof. Dr. Bernd-Christian Renner.
- Replaced frequency shift keying with **chirp modulation** for packet-based preamble synchronisation between micro autonomous underwater vehicles (μ AUVs), where multipath propagation was making FSK preamble boundaries ambiguous.
- Built the signal generation algorithm and evaluation chain, and characterised the acoustic modem and Tiepie Handyscope HS5 instrumentation to select a workable parameter set.
- Validated across **four environments** of increasing realism — small container, test tank, Channel Harburg, and Außenmühle Lake — rather than in simulation alone.
- Showed chirp yields a prominent correlation peak where FSK produces a multipath-distorted triangular region, reducing preamble detection delay from **seconds to milliseconds** across ten iterations, with robustness to both multipath and Doppler shift.
- Verified the proportionality between **time-bandwidth product** and main-lobe power concentration, and established that the generation sampling rate could be reduced from 2 MHz to 500 kHz before spurious correlation peaks appeared — the constraint for porting to embedded hardware.

Selected Activities

Nanotechnology Solutions in Health — NanoMed concept study

Barcelona, Spain / Bydgoszcz, Poland

ERASMUS+ COLIBRI SEMINAR — judged best project

Apr. 2016 - Jul. 2016

- Feasibility study for continuous monitoring of cancer treatment response: surveyed the enabling technologies and proposed a concept architecture of implantable biosensor motes, a dissolvable relay node, a wearable gateway and cloud reporting to clinicians.
- Assessed candidate biosensor modalities against measurable parameters, reviewed WBAN and Bluetooth as the in-body link, and mapped the EU medical device route to market (CE marking, MDD, AIMD, clinical phases).
- Built the business model canvas for the concept. Supervised by Mehmet Şükrü Kuran and Tuna Tugcu; judged best project among the participating international teams.

Institute of Engineering and Management

Kolkata, India

BACHELOR PROJECT

Oct. 2013 - May 2014

- *Pongopal*: Arduino-based floor-cleaning robot evaluating real-time path minimisation approaches. Grade 10/10.

Publications & Presentations

aMP: Deep learning framework to characterize meiosis progression timeline in tetraploid *Arabidopsis thaliana*

Universität Hamburg

DOCTORAL DISSERTATION

2024

- Published open access. URN [urn:nbn:de:gbv:18-ediss-123289](https://nbn-resolving.org/urn:nbn:de:gbv:18-ediss-123289) — ediss.sub.uni-hamburg.de/handle/ediss/11301

International Congress on Sexual Plant Reproduction

Chengdu, China

POSTER & FLASH TALK (selected) — ICSPR 2026

Sep. 2026

- *JASON acts as a molecular glue mediating the clustering of meiotic organelles ensuring the fidelity of chromosome segregation*.
- Y. Hamamura, A. Ghosh, N. De Winne, G. De Jaeger, M. Sato, M. Wakazaki, K. Toyooka, S. Komaki, A. Schnittger.
- Max Planck Institute of Molecular Plant Physiology, Universität Hamburg, Ghent University, VIB, RIKEN CSRS, NAIST.

Plant Meiosis Meeting

Poznań, Poland

POSTER

Sep. 2024

- *Automated clustering techniques for meiosis staging in Arabidopsis thaliana*.

International Conference on Arabidopsis Research

POSTER — ICAR 2023

Chiba, Japan

Jun. 2023

- *Cell-cycle synchronized organelle clustering in meiocytes.*
- Y. Hamamura, S. Komaki, N. De Winne, G. De Jaeger, **A. Ghosh**, A. Schnittger.
- Universität Hamburg, NAIST, Ghent University, VIB Center for Plant Systems Biology.

Microtubules: From Atoms to Complex Systems

DIGITAL POSTER — EMBO|EMBL SYMPOSIUM

Heidelberg, Germany

Jun. 2020

- Microtubule structures revealed as the main feature during deep-learning-based auto-staging of plant meiosis.